KOUROVKA NOTEBOOK PROBLEM 21.15

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21.15. Suppose B is a subgroup of the symmetric group S_Ω on an infinite set Ω. Will the amalgamated free product $S_\Omega *_B S_\Omega$ of two copies of S_Ω with amalgamation of B be embeddable in S_Ω? This is a weakened form of the group case of Question 4.4 in (G. M. Bergman, *Indag. Math.*, **18** (2007), 349–403).

It is known that $S_\Omega *_B S_\Omega$ need not be so embeddable by a map respecting B (*Algebra Number Theory*, **3** (2009), 847–879, § 10).

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The answer to Kourovka Notebook Problem 21.15 seems to be "Not necessarily". Take Ω countably infinite, B a subgroup of S_Ω which is not Borel, and $\phi : S_\Omega *_B S_\Omega \to S_\Omega$ an injective group homomorphism. Let ϕ_1 be the restriction of ϕ to the first copy of S_Ω and ϕ_2 that to the second. Each ϕ_i is continuous [2], so $\mathrm{im}(\phi_i)$ is Borel in S_Ω (as a continuous injective image of a Polish space). Now $B = \phi_1^{-1}(\phi(B)) = \phi_1^{-1}(\mathrm{im}(\phi_1) \cap \mathrm{im}(\phi_2))$ is Borel, a contradiction.

A longer proof with all of the gory details. With Ω countably infinite, the group S_Ω is a Polish group under the natural topology. Take $B \leq S_\Omega$ to be a subgroup which is not Borel in S_Ω. Such a B is easily found, for example by using countably infinitely many disjoint transpositions in S_Ω to find a copy of the compact group $\prod_{\mathbb{N}} \mathbb{Z}/2\mathbb{Z}$ in S_Ω, and a non-Borel subgroup of $\prod_{\mathbb{N}} \mathbb{Z}/2\mathbb{Z}$ (hence of S_Ω) is given by the sequences which are in the complement of a given nonprincipal ultrafilter on $\mathbb{N}$. Now suppose that $\phi : S_\Omega *_B S_\Omega \to S_\Omega$ is an abstract injective group homomorphism. Let ϕ_1 be the restriction of ϕ to the first copy of S_Ω in the amalgamated product $S_\Omega *_B S_\Omega$ and ϕ_2 the restriction of ϕ to the second copy. Note that each of these two copies of S_Ω has a natural (and unique) Polish group topology, but we are not going to consider any topology on the amalgamated free product $S_\Omega *_B S_\Omega$.

Since S_Ω is well-known to have ample generics, by [2, Theorem 1.10] any homomorphism from S_Ω to a Polish group is continuous, and therefore each ϕ_i is continuous. Then the image $\mathrm{im}(\phi_i)$ is Borel in S_Ω by the classical fact that the continuous injective image of a Polish space is Borel (see [1, Theorem 15.1]). Therefore $\mathrm{im}(\phi_1) \cap \mathrm{im}(\phi_2)$ is Borel, and we have $B = \phi_1^{-1}(\phi(B)) = \phi_1^{-1}(\mathrm{im}(\phi_1) \cap \mathrm{im}(\phi_2))$ is Borel (in the first copy of S_Ω), contrary to how B was chosen.

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References

[1] A. S. Kechris, Classical Descriptive Set Theory, Springer, Berlin, 1995.

[2] A. S. Kechris, C. Rosendal, *Turbulance, amalgamation, and generic automorphisms of homogeneous structures*, Proc. Lond. Math. Soc. 94 (2007), 302-350.